Infrastructure

Week 9

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Today

Week 8 ended with Troesken's claim that waterworks were among the largest public investments in American history. This week asks what an investment of that kind is worth to the economy as a whole.

1. **A frame.** Two reasons a market price understates the social value of infrastructure, an externality and a wedge.

2. **A fire.** Hornbeck and Keniston on the Great Boston Fire of 1872, which destroyed 776 buildings and left the land under them worth more than before.

3. **A network.** Hornbeck and Rotemberg on the railroads. Counting the economy's distortions roughly triples Fogel's estimate of what they were worth.

What is a public investment worth and how would anyone find out?

What infrastructure is worth

Two returns to the same investment

- Every public investment has two returns. The **private return** is what it pays whoever built it. The **social return** is what it pays everyone.
- The railroads of 1890 cost about $8 billion and their owners earned about 3.5 percent a year on it.
- Hornbeck and Rotemberg put the social return at 48 percent a year. The number comes from a model that removes the railroads and lets workers leave the country, which this deck reaches at the end. Investors collected about 7 percent of what the network was worth to the economy.

Hornbeck and Rotemberg (2024), p. 3585 and n. 17. The cost of the network is Adams's (1895) estimate.

▸ The arithmetic

Two reasons a market price understates social value

Externality

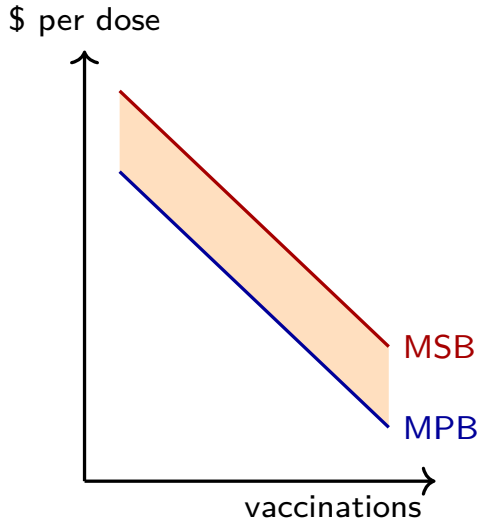
Part of the benefit of an action lands on people who did not pay for it. The person deciding counts too little of the benefit and does too little. Week 8's vaccination was the example.

Wedge

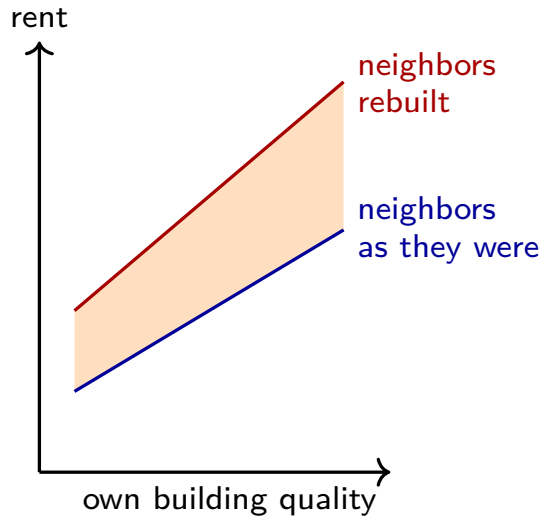
The last unit of an input a firm uses, a worker or a ton of coal, adds more to revenue than it costs. A firm that cannot borrow enough, or that holds its price up to protect a markup, stops short of the point where the two are equal.

- Boston is a case of the first. The railroads are a case of the second.
- Week 5 met wedges through Hsieh and Klenow, whose Indian and Chinese plants with the same technology earned very different revenue per unit of input because some faced distortions others did not. This deck writes a wedge as ψ, following Hornbeck and Rotemberg.

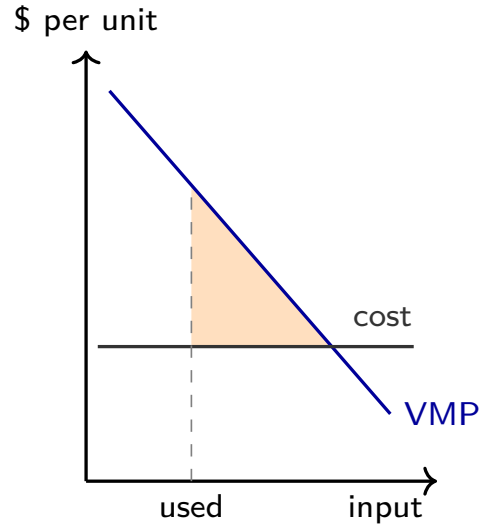
Uncounted value in three settings



Week 8, benefit to people not vaccinated



Boston, rent added by the neighbors' rebuilding



Railroads, value of inputs firms stop short of

In each panel the shaded area is value the person deciding does not count. In the first two it goes to someone else. In the third it is lost because firms stop short of the efficient amount. A fire that makes every owner rebuild at once, or a railroad that lets firms sell more, expands the activity and recovers part of the area.

Stylized diagrams. VMP is the value of the marginal product, MSB and MPB marginal social and private benefit.

Hornbeck and Keniston on the Great Boston Fire

Boston, 9–10 November 1872



The ruins seen from opposite Trinity Church on Summer Street. Frank Leslie's Publishing House, 1872, from photographs by J. W. Black. Library of Congress, public domain.

- The alarm was late and an illness among the fire department's horses slowed the engines. The fire burned for 22 hours until engines arrived from surrounding towns.
- It destroyed 776 buildings in the business district, about $75 million in damages, or 11 percent of the value the city's tax assessors put on all the real estate and personal property in Boston. Insurers paid roughly half.

Hornbeck and Keniston (2017), p. 1369 and n. 2.

The burned area on the 1867 map of downtown Boston



- Red is the burned area, a dense commercial district next to the center of the city.
- Each black dot is a plot the authors located on the 1867 map.
- The pink and blue tints are the original map's own shading.

Hornbeck and Keniston (2017), Figure 1, on the Sanborn Map Company's 1867 map, from the working-paper version, which reproduces the map more sharply. The map has no scale bar.

Who rebuilt the burned district

- Private owners rebuilt it plot by plot. Boston had no zoning and the building code passed after the fire was weak and largely rescinded in 1873.
- The city spent $5.7 million to widen and extend some streets and to create Post Office Square. Landowners blocked a larger reorganization of the street plan.
- Construction technology did not yet allow taller buildings. Owners rebuilt to a higher quality at the same height.
- Boston real estate had been growing 4.8 percent a year in real terms for 20 years. In Week 6's terms this is a shock to a growing city. In a declining city the paper expects the same forced rebuilding to bring economic losses.

Hornbeck and Keniston (2017), pp. 1367, 1369–1370, 1388–1389, and 1394.

The question the paper asks

- Week 6 cited this paper in one line as evidence that proximity pays.
- It looks for **spillovers**, effects of one building on the value of the buildings beside it, that run over a few blocks rather than a whole city.
- It writes down two models, with and without those spillovers, finds the predictions on which they disagree, and uses the fire to test them.

The test turns on land. In a model with spillovers a fire can raise the value of the land it burns. In a model without them it cannot.

The baseline model has no spillovers

- A building of quality q earns rent $r(q, \omega)$, where ω is the productivity of the city, which grows over time.
- A building cannot be improved a bit at a time. Changing q means paying for a whole new building. The best quality to build rises with ω.

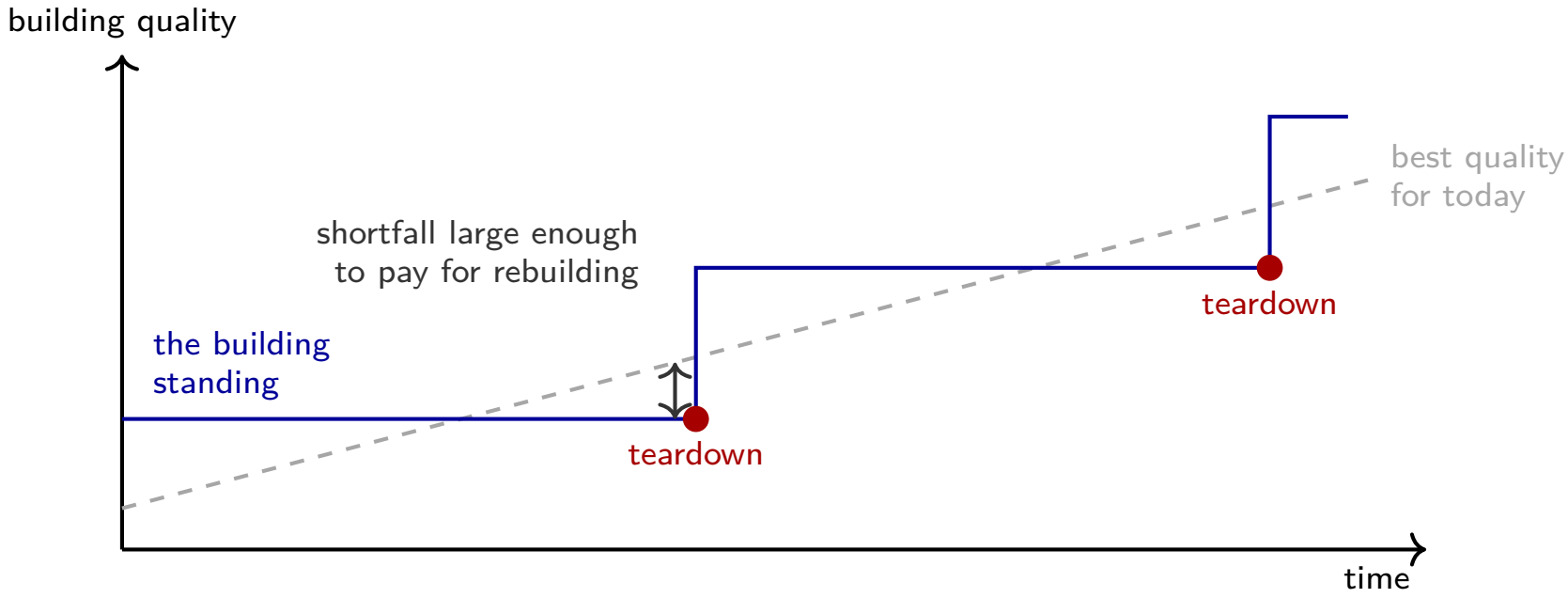
Land value and building value

An **option value** is what the right to act later is worth today. Write $V(q, \omega)$ for what a plot carrying a building of quality q would sell for. The land is worth what the plot would sell for empty, $V(0, \omega)$, which is the option value of building on it well from now on. The building is worth the rest, $V(q, \omega) - V(0, \omega)$.

- A building ready to be torn down adds nothing. A plot carrying one is worth exactly its land.
- If your building burns and you rebuild, you replace an aging building with one built for today, so your building value rises. The empty plot is worth what it was, so your land value does not move.

Hornbeck and Keniston (2017), pp. 1373–1375.

Owners replace buildings on an (s,S) rule



- An **(s,S) rule**, named for inventory models that restock up to S once stock falls to s, waits until a shortfall reaches a trigger and then closes it in one jump. Owners overbuild for today, hold the building while it becomes obsolete, and rebuild when the shortfall covers the cost.
- Each owner's clock started on a different date so a city block holds buildings of many vintages.

Stylized. Hornbeck and Keniston (2017), pp. 1373–1375, where owners "over-build for contemporaneous conditions."

The extended model adds two spillovers

Rent now depends on Q, the average quality of the buildings nearby, as well as on the building's own quality, $r(q, Q, \omega)$.

The two assumptions

1. Rent rises with the neighbors' quality, $\partial r/\partial Q > 0$. A plot next to good buildings earns more.
2. The return to your own quality rises with the neighbors' quality, $\partial^2 r/\partial q\,\partial Q > 0$. Good neighbors make it worth building better yourself.

- Boston's downtown had hundreds of small owners and no contract that bound them to rebuild together. Each rebuilt on its own clock and the block never upgraded all at once.
- The fire set every clock in the burned area to zero on the same night.

Hornbeck and Keniston (2017), p. 1374. In the paper's simulations η is the weight of Q in rent, and $\eta = 0$ is the baseline model.

Predictions of the two models

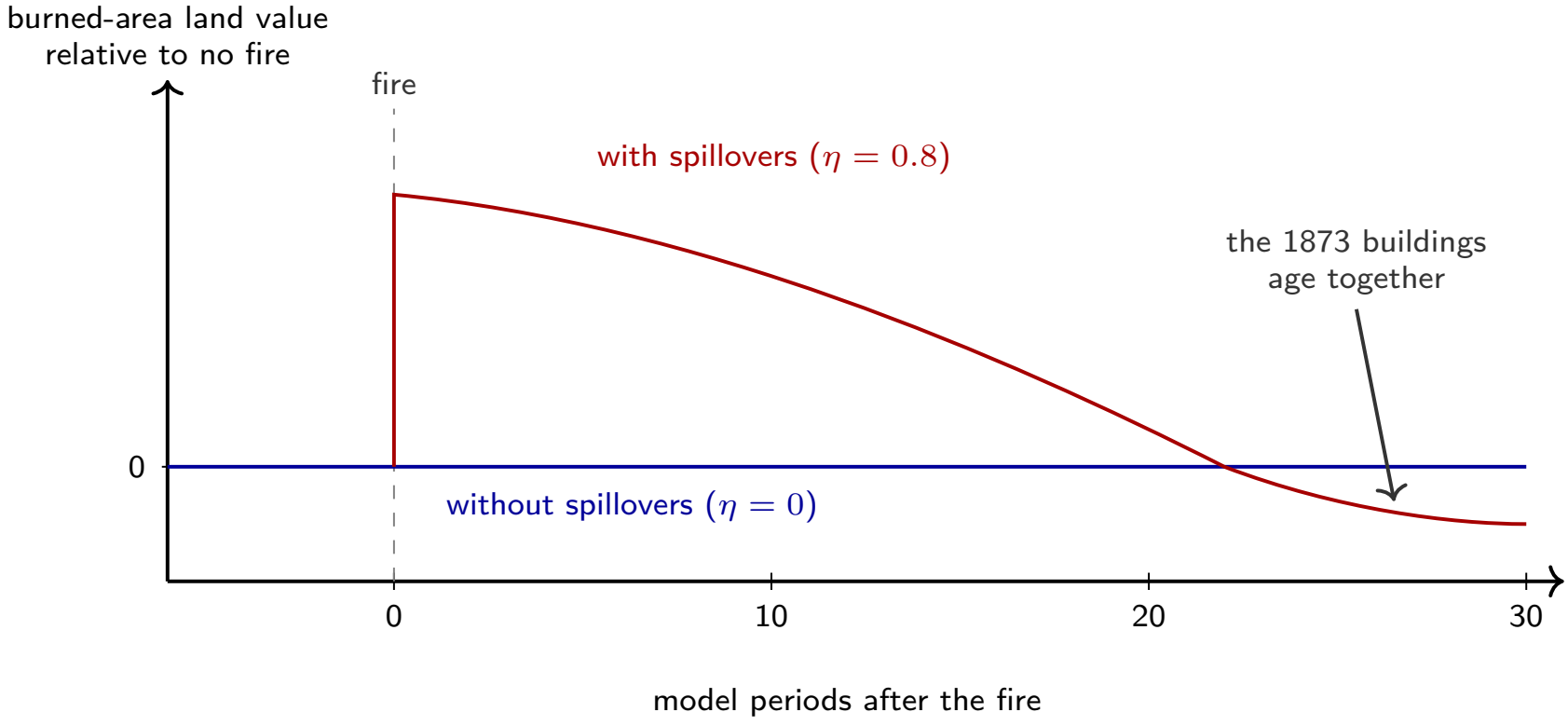
Without spillovers a fire resets the clocks on the burned plots and does nothing else. The new buildings are worth more than the old ones and the land under them is worth what it was. With spillovers, the fire…

1. increases plot land values in the burned area and the gain fades over time, perhaps below land values in unburned areas.
2. increases land values in nearby unburned areas.
3. increases average building values in the burned area, following reconstruction.
4. has an impact on building values that is largest for the cheapest buildings but present even for the most valuable ones.
5. increases building values in nearby unburned areas.
6. has a greater impact on building values than individual fires.

Unshaded text is what both models predict. Shading marks where the extended model departs from the baseline.

Hornbeck and Keniston (2017), pp. 1375–1378, where the paper groups these as four predictions. The baseline model implies the opposite of each shaded clause.

What the fire does to land value in the model

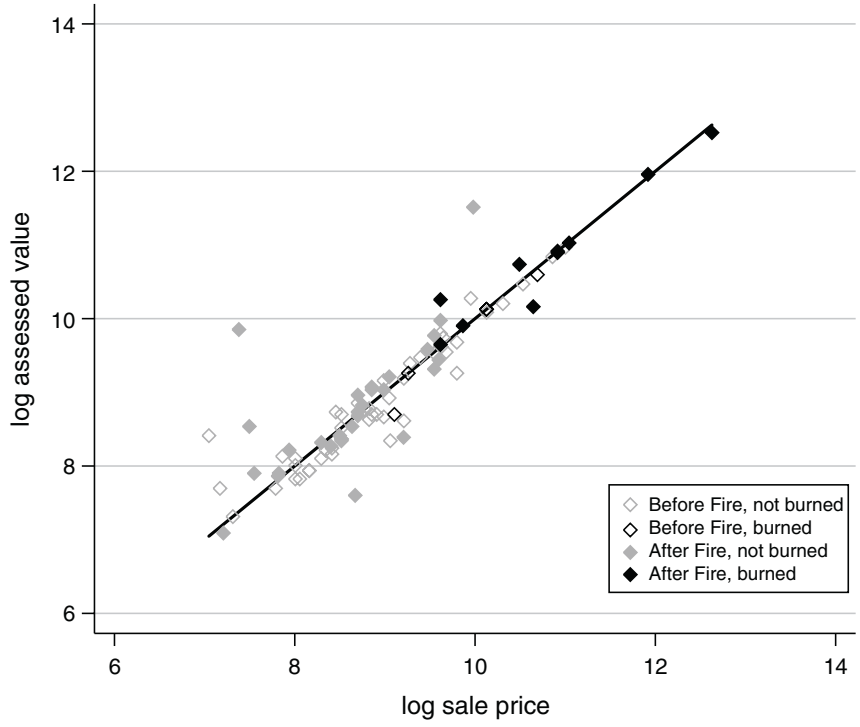


- With spillovers the land jumps at the fire because every neighbor is about to rebuild at once. The gain fades as the new buildings age.
- Land value can then fall below the no-fire path because a whole district of buildings from the same year becomes obsolete together.

Stylized redrawing of the land-value panel of Hornbeck and Keniston (2017), Figure 3, from the paper's simulation.

The assessment data

- Boston's assessors valued land and building separately on every plot every year. The authors digitized the ledgers for 1867, 1869, 1871, 1872 (before the fire), 1873, 1882, and 1894, a total of 44,543 plot-years.
- Plots are placed on fire-insurance maps. Sales come from the Registry of Deeds and 109 individual building fires from fire department records.



Log assessed value against log sale price, 88 plots. The markers separate burned from unburned plots and sales before the fire from sales after it. Hornbeck and Keniston (2017), Figure 2.

Hornbeck and Keniston (2017), pp. 1370–1371 and 1379.

The design in cells

Two years, one before the fire and one after.

$$\ln V_{it} = \alpha + \beta_1 \, FireArea_{it} + \beta_2 \, PostFire_{it} + \beta_3 \, FireArea_{it} \times PostFire_{it} + \beta_4' X_{it} + e_{it}$$

Land value

	Before the fire	After the fire
Unburned area	α	$\alpha + \beta_2$
Burned area	$\alpha + \beta_1$	$\alpha + \beta_1 + \beta_2 + \beta_3$

How much does land value rise in the unburned area? β_2
How much does land value rise in the burned area? $\beta_2 + \beta_3$

So β_3 is the effect of the fire on land value in the burned area relative to the unburned area. This is Week 2's difference in differences.

Identifying assumption, in their words

"Plots in the burned area would have changed the same as plots in the unburned area, on average, in the absence of the Fire. This identification assumption is more plausible over shorter periods of time, and becomes more demanding over longer periods of time after the Fire."

X_{it} holds each plot's pre-fire values. The paper estimates a β_3 for each of 1873, 1882, and 1894 against 1872. Hornbeck and Keniston (2017), equations (1) and (2), pp. 1378–1379.

The second test replaces burned with distance

- Replace the burned indicator with indicators for 100-foot bands of distance from the fire's edge. The comparison group is plots more than 2,900 feet away.
- A second version lets the effect fall in a straight line outside the burned area and estimates the distance at which it reaches zero.
- Burned plots were worth far more before the fire, 0.772 log points more per square foot in 1872, more than double (a difference of x log points is a ratio of e^x). The gap did not change from 1871 to 1872 (-0.010, SE 0.012), so the two groups were on parallel paths before the fire.
- So the paper also splits the unburned plots into a **close** sample, more like the burned plots but lifted by any spillover, and a **distant** sample, untouched by the spillover but less alike. Neither is clean.

Hornbeck and Keniston (2017), pp. 1372–1373 and 1379–1380, Table 1.

What the fire did

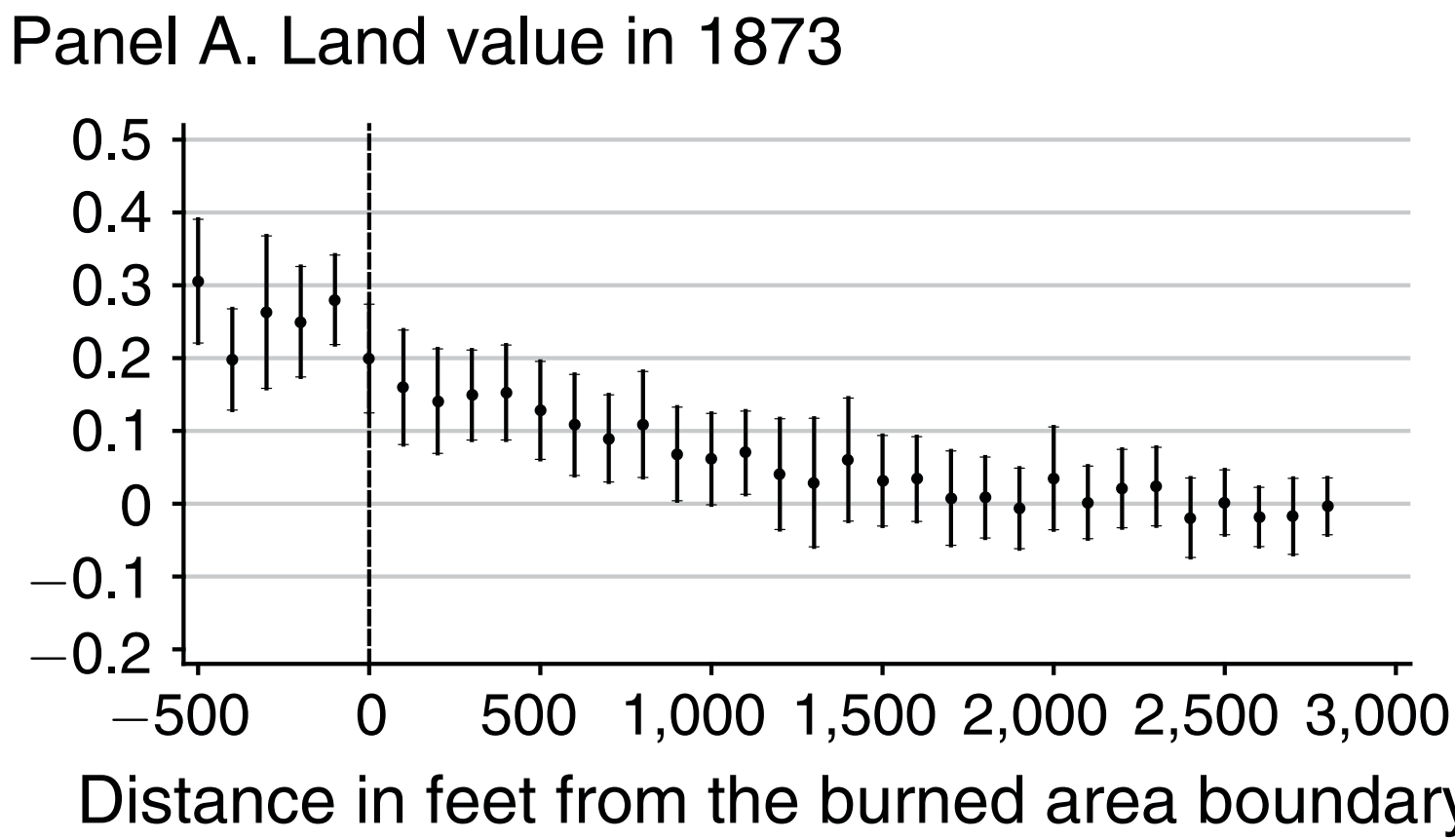
Burned land rose about 15 percent in a year

Log land value per square foot	Full sample (a)	(b)	Close (c)	Distant (d)
1873 × burned	0.151	0.167	0.147	0.261
	(0.019)	(0.019)	(0.020)	(0.031)
1882 × burned	0.160	0.143	0.130	0.284
	(0.043)	(0.038)	(0.037)	(0.065)
1894 × burned	−0.100	−0.140	−0.149	−0.143
	(0.056)	(0.058)	(0.060)	(0.077)
Pre-fire value controls		yes	yes	yes
Plot-years	44,543	44,543	19,944	28,438

- An estimate of 0.151 log points is 16.3 percent, which the paper rounds to "roughly 15 percent." The gain held through 1882 and had reversed by 1894, which the paper reads as the model's cohort effect, a district of same-year buildings aging together. Boston's first skyscrapers went up just outside the burned area in those years, and assessments that revert to their mean would look the same.
- The distant sample's estimate is larger because its comparison group leaves out the nearby unburned plots, whose land also rose. In the full sample those plots sit in the comparison group and pull the estimate down. Prediction 1 of the extended model holds and the baseline's fails.

Hornbeck and Keniston (2017), Table 2, p. 1380. Standard errors, clustered by block, in parentheses.

Unburned land near the fire gained almost as much



Land just outside the burned area gained about as much as land inside it and the gain fell to zero at an estimated 1,339 feet (SE 112), about 25–30 buildings away. Prediction 2 holds. By 1882 buildings on nearby unburned plots had also been rebuilt better, with the gain falling in distance (Figure 4, panel D). Prediction 5 holds.

Change in log land value 1872–1873 by 100-foot band from the fire's edge, relative to plots beyond 2,900 feet. Negative distances are inside the burned area. No such pattern existed before the fire. Hornbeck and Keniston (2017), Figure 4, panel A.

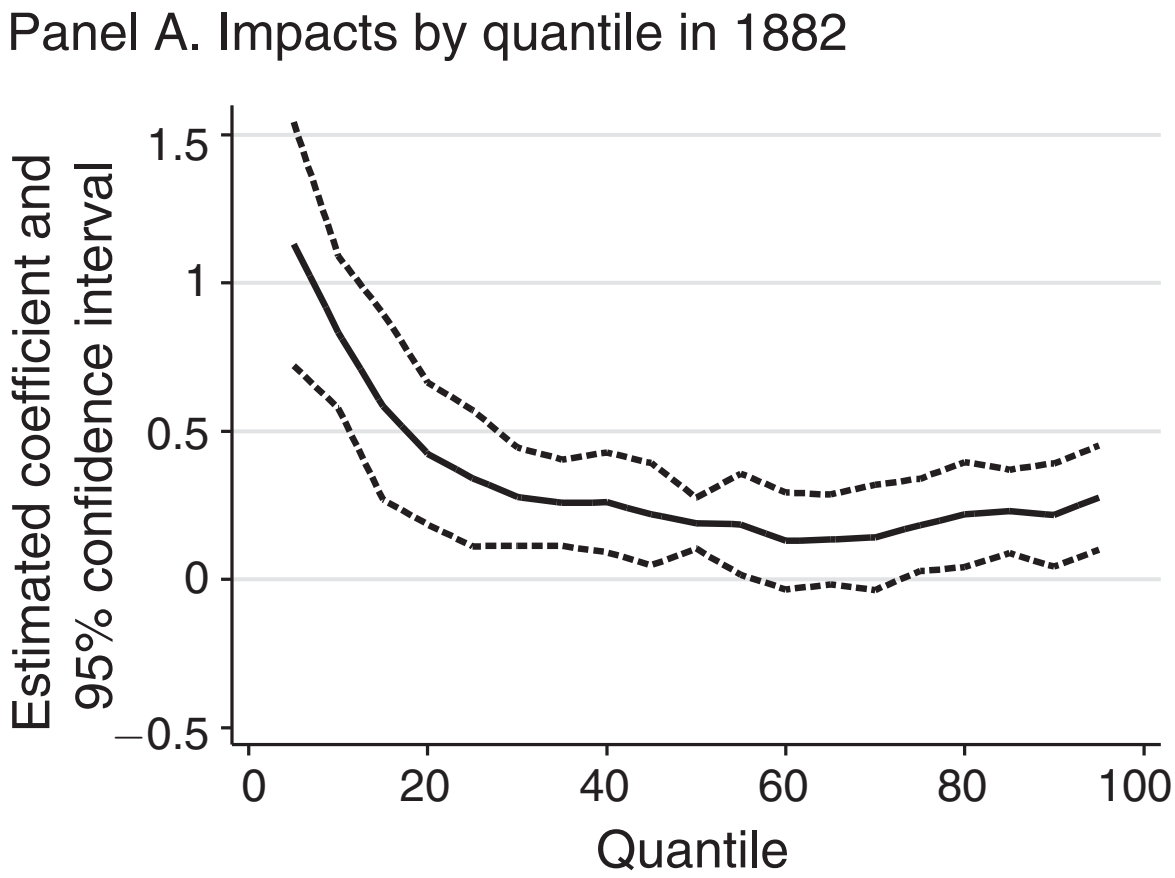
Building values rose in the burned area

	(1)	(2)
1873 × burned	−1.800	−2.024
	(0.161)	(0.168)
1882 × burned	0.404	0.509
	(0.068)	(0.053)
1894 × burned	0.177	0.399
	(0.078)	(0.079)
Pre-fire controls		yes

- Log building value per square foot, burned plots relative to unburned.
- In 1873 the burned area was being rebuilt so building value collapsed.
- By 1882 burned plots' buildings were worth 0.404 log points more, about 50 percent, and 0.509 with pre-fire controls. Prediction 3, which both models share, holds.
- The gain had shrunk by 1894 as the new buildings aged.

Hornbeck and Keniston (2017), Table 4, p. 1384. Many 1873 buildings were assessed partly built.

The gain reached the top of the distribution



By 1882 the gain is about 1 log point, nearly a tripling, for the cheapest buildings and about 0.2 for the most valuable, and the 0.2 is still significant. Without spillovers the most valuable buildings would not move. Prediction 4 holds.

Effect of the fire on log building value by quantile of the building-value distribution, cheapest at left, burned relative to unburned, 1882, with 95 percent bands. By 1894 all but the lowest quantiles had converged (panel B). Hornbeck and Keniston (2017), Figure 6, panel A.

The Great Fire against 109 individual fires

	Log land value	Log building value
Great Fire, 1873	0.163 (0.019)	-1.960 (0.174)
Great Fire, 1882	0.141 (0.039)	0.514 (0.056)
Great Fire, 1894	-0.150 (0.057)	0.401 (0.082)
Individual fire, about 1 year after	-0.039 (0.050)	-0.135 (0.171)
Individual fire, about 10 years after	0.089 (0.106)	0.344 (0.154)
Individual fire, about 22 years after	-0.214 (0.274)	0.014 (0.087)
p, equal at about 1 year	0.000	0.000
p, equal at about 10 years	0.647	0.295
p, equal at about 22 years	0.822	0.000

- A single burned building leaves its own land value unchanged, as both models predict. The Great Fire raised it.
- At 10 years individual fires raised building value by 0.344 against the Great Fire's 0.514 ($p = 0.295$ for equality). By 22 years their effect was gone and the Great Fire's remained. Prediction 6 holds.

Hornbeck and Keniston (2017), Table 5, p. 1388. The sample is smaller than in Table 2, so the Great Fire rows differ slightly.

The scale of the fire was the mechanism

One owner who rebuilds after a fire gets a better building on the same land. Hundreds of owners who rebuild in the same year raise each other's rents so the land under all of them is worth more. In the model the individual fires and the Great Fire differ only in how many neighbors rebuilt at once.

- This is Week 8's externality in a new place. Each owner's rebuilding raised the neighbors' rents and no owner was paid for it.
- The fire made coordination unnecessary for a few years by forcing everyone to rebuild together.

Hornbeck and Keniston (2017), pp. 1386–1388 and 1396. In the data individual fires also skew toward more valuable buildings (n. 26).

Other ways a fire could raise land values

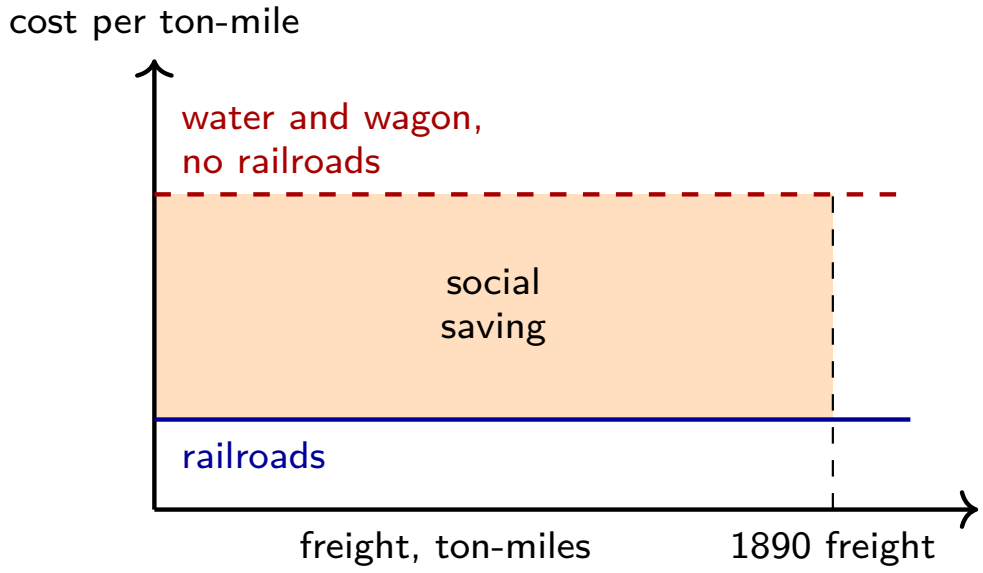
Alternative	**What the paper finds**
Better public works (wider streets, water and sewer pipes)	$5.7 million of street changes and $85,000 on water mains. Controls for street width, pipe size, and distance to Post Office Square leave the estimates in place. The 1869–1870 street widening around Washington Square is the comparison.
Rationalization (a blank slate lets uses sort sensibly)	Industries became no more clustered by Ripley's L, a measure of spatial clustering. Plots grew 6–16 percent, enough to explain 1–2 percent of land value.
Ownership consolidation	Only two landowners held three or more additional plots in 1873.

What remains after these is the spillover from the neighbors' buildings.

Hornbeck and Keniston (2017), pp. 1388–1392, Tables 6 and 7.

Fogel, market access, and the railroads

Fogel's social saving



- The **social saving** is the extra cost of carrying the goods the railroads carried in 1890 by water and wagon instead, with everything else held fixed.
- Fogel (1964) put it at 2.7 percent of GNP. Economists before him had called the railroad indispensable.
- In the network Hornbeck and Rotemberg use, rail cost 0.63 cents per ton-mile and water 0.49. Going without rail meant wagon hauls at 23.1 cents and 50 cents a ton for each change of carrier.

Stylized. Rates from Donaldson and Hornbeck, as used by Hornbeck and Rotemberg (2024), n. 4, p. 3556.

Why 2.7 percent is not an upper bound

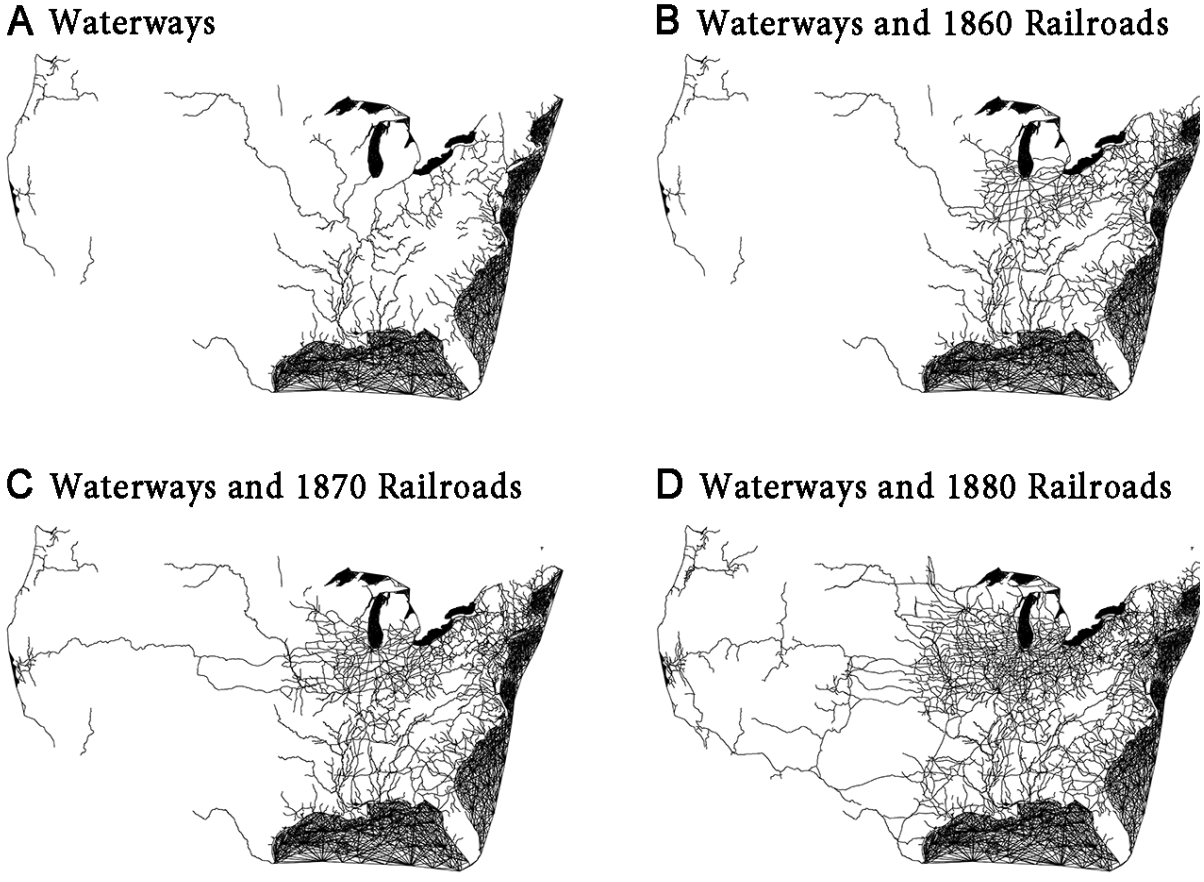
Fogel's rectangle rests on three assumptions.

1. The freight shipped stays at its 1890 level. Fogel argued this made 2.7 an upper bound since shippers facing higher costs would ship less.
2. The rest of the economy keeps its shape, where people live and what they make.
3. Inputs are allocated efficiently. Moving them from one place to another creates no value of its own.

- Donaldson and Hornbeck (2016) drop the second. In a model where people and trade move, the railroads' value shows up in the price of farmland, and the fall in farmland value without them is 3.2 percent of GNP.
- Hornbeck and Rotemberg drop the third. If other activities earn more than they cost, the social saving "is not an upper bound for the welfare effects of the railroad network."

Hornbeck and Rotemberg (2024), pp. 3548–3551 and 3591. Week 6's Bleakley and Lin studied the same railroads making portage sites obsolete.

The network grew from waterways to railroads



By 1880 railroads reached most of the counties that rivers and canals had left to wagons.

The transport network used to compute market access. Waterways alone (A), then with the railroads built by 1860 (B), 1870 (C), and 1880 (D). Hornbeck and Rotemberg (2024), Figure 2.

Market access

A county's market access is one number for how much of the country's population it can reach.

Market access

$$MA_c = \sum_{d \neq c} \tau_{cd}^{-\theta} L_d$$

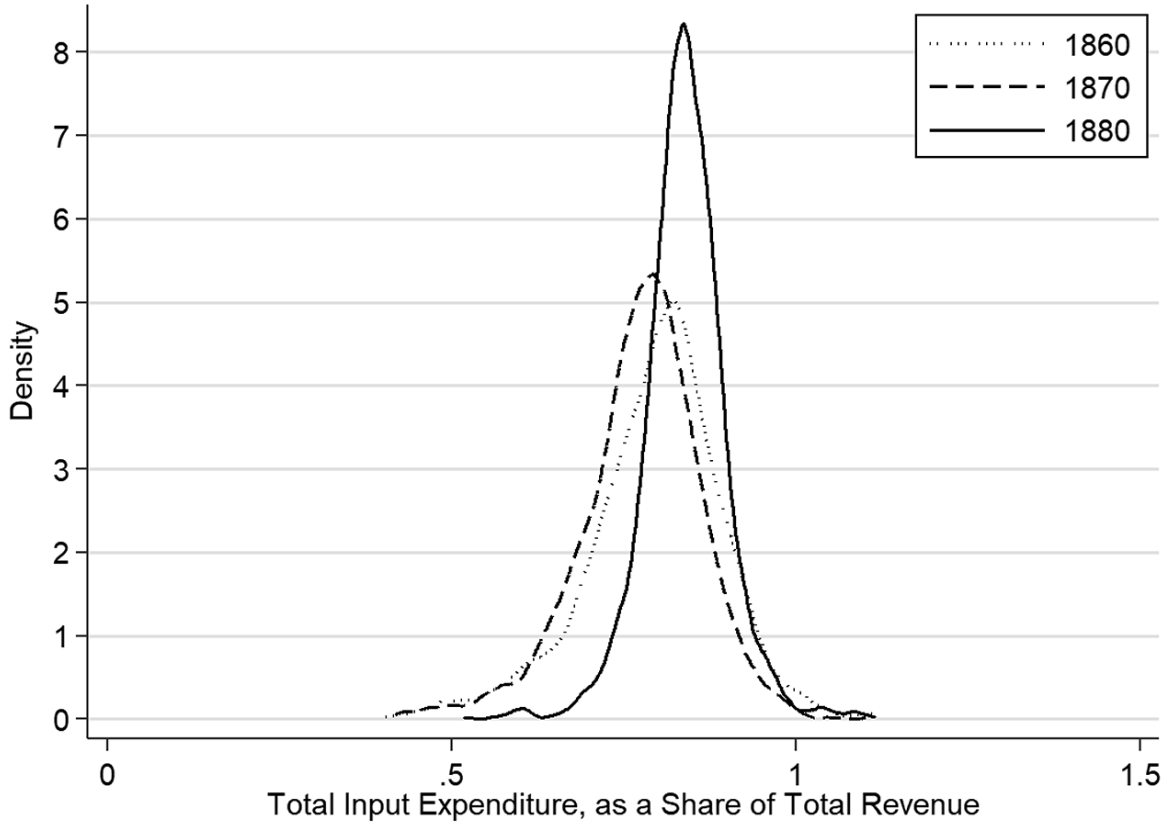
County c's market access adds up the population L_d of every other county, discounted by how costly it is to ship there.

- Trade elasticity. $\theta = 3.05$ sets how fast a market's weight falls as the cost of reaching it rises. The higher is θ, the more responsive is trade to an increase in travel cost.
- t_{cd} is the freight charge along the cheapest route over rivers, lakes, canals, railroads, and wagon roads. A county gains market access when a railroad replaces a long wagon haul.

Hornbeck and Rotemberg (2024), pp. 3556–3557.

Hornbeck and Rotemberg on the railroads

Manufacturers spent about 80 cents per dollar of revenue



If doubling every input doubled output and each input were paid what it adds, paying for inputs would use up all of revenue. In the typical county spending was 0.8 of revenue, so a fifth of revenue was left over.

County spending on materials, labor, and capital as a share of manufacturing revenue in the censuses of 1860, 1870, and 1880. The mean is 0.8 (SD 0.1) and 1.5 percent of counties exceed 1. Hornbeck and Rotemberg (2024), Figure 1 and n. 2.

The wedge and the gap

The wedge

Let α^k be the percent rise in output from a 1 percent rise in input k, its output elasticity, R revenue, and $E^k = W^k X^k$ spending on input k, price times quantity. A firm facing wedge ψ^k spends

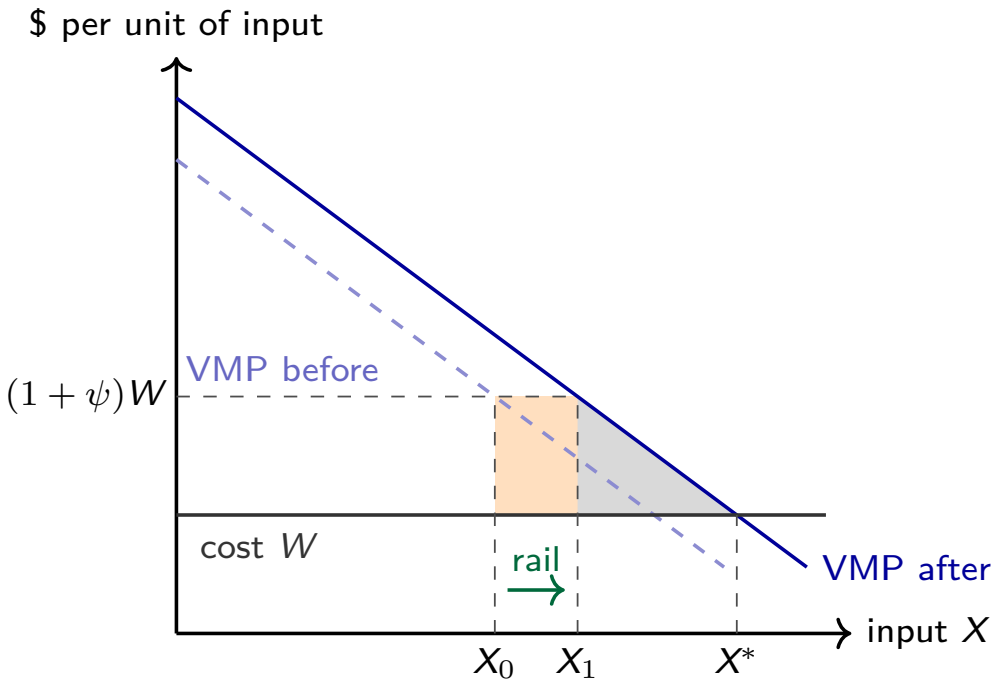
$$(1 + \psi^k)\, E^k \;=\; \alpha^k R,$$

so input k's share of revenue is $s^k = \alpha^k/(1 + \psi^k)$, below α^k, and the wedge is $\psi^k = (\alpha^k - s^k)/s^k$. A marginal dollar spent on input k returns $1 + \psi^k$ dollars of revenue.

- Raising spending on k by 1 percent raises revenue by α^k percent and costs s^k percent of revenue, so the **gap** $\alpha^k - s^k$ is the surplus from that 1 percent, as a share of revenue.
- In 1860 the elasticities were 0.71 for materials, 0.25 for labor, and 0.04 for capital, which sum to one. Wedges ran from roughly a fifth to a third and the gap was largest for materials.
- A persistent surplus is more than ordinary profit. With constant returns, paying every input what it adds would use up revenue, so revenue left over means some input is being paid less than it adds.

Hornbeck and Rotemberg (2024), pp. 3553–3555 and equation (9), p. 3575.

A railroad moves the firm toward the efficient point



- With the wedge the firm stops where the value of the marginal product (VMP) equals $(1+\psi)W$, at X_0.
- Better market access lets the firm sell its output in more places at better prices, so each unit of input produces output worth more and VMP shifts up. The firm expands to X_1. Each added unit is worth $(1+\psi)W$ and costs W. The orange rectangle is new surplus with no change in technology, which the paper calls allocative efficiency.
- Without the wedge the firm would go on to X^*. The gray triangle is what the wedge still costs.

Stylized, with the wedge held fixed as in the paper.

▸ The decomposition

The design

$$Y_{ct} = \beta \ln MA_{ct} + \gamma_c + \gamma_{jt} + \gamma_t f(\text{latitude}_c, \text{longitude}_c) + \varepsilon_{ct}$$

- c is a county, j its state, t the census year, 1860, 1870, or 1880. County effects remove anything fixed about a county. State-by-year effects remove anything that hit a whole state in a decade. f is a smooth function of the county's location whose effect can change each decade, so each county is compared with nearby counties.
- Outcomes are log revenue, log spending on each input, log productivity, TFPR, and AE. Standard errors are clustered by state.
- Coefficients are scaled to a 1-standard-deviation larger gain in market access from 1860 to 1880, which is a 24 percent larger gain.

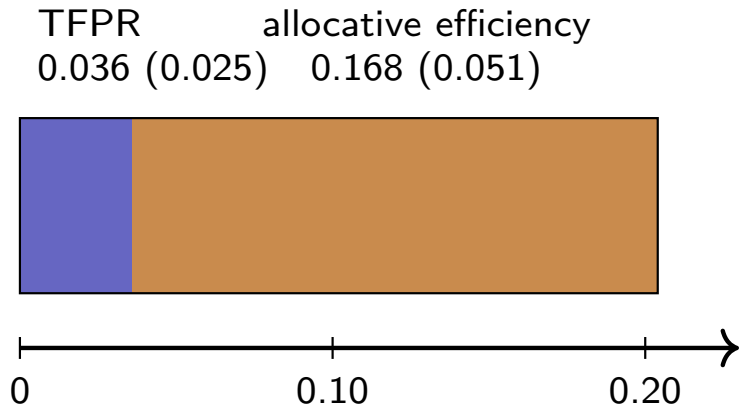
Identifying assumption, in their words

"Counties with relative increases in market access would otherwise have changed similarly to nearby counties."

Hornbeck and Rotemberg (2024), equation (4), p. 3561, and p. 3562. The design is Donaldson and Hornbeck's (2016).

Market access raised revenue, every input, and surplus

Outcome	Effect of 1 SD
Log revenue	0.192 (0.049)
Log capital spending	0.158 (0.053)
Log labor spending	0.196 (0.061)
Log materials spending	0.183 (0.050)
Log surplus	0.204 (0.051)
TFPR	0.036 (0.025)
AE	0.168 (0.051)



log surplus, per 1 SD of market access

- Revenue and spending on each input rose by 0.16–0.20 log points and surplus by 0.20.
- Of the 0.204, 0.168 came from more inputs at the existing gaps and 0.036 from TFPR.
- The decomposition checks out. With $\nu \approx 5.1$, gaps summing to about 0.2, and inputs up about 0.18, AE should be about $5.1 \times 0.2 \times 0.18 \approx 0.18$, close to the estimated 0.168.

Hornbeck and Rotemberg (2024), Tables 1 and 2, column 1, pp. 3563–3564, 1,802 counties and 5,406 county-years. The paper calls surplus county productivity.

The expansion came from new establishments

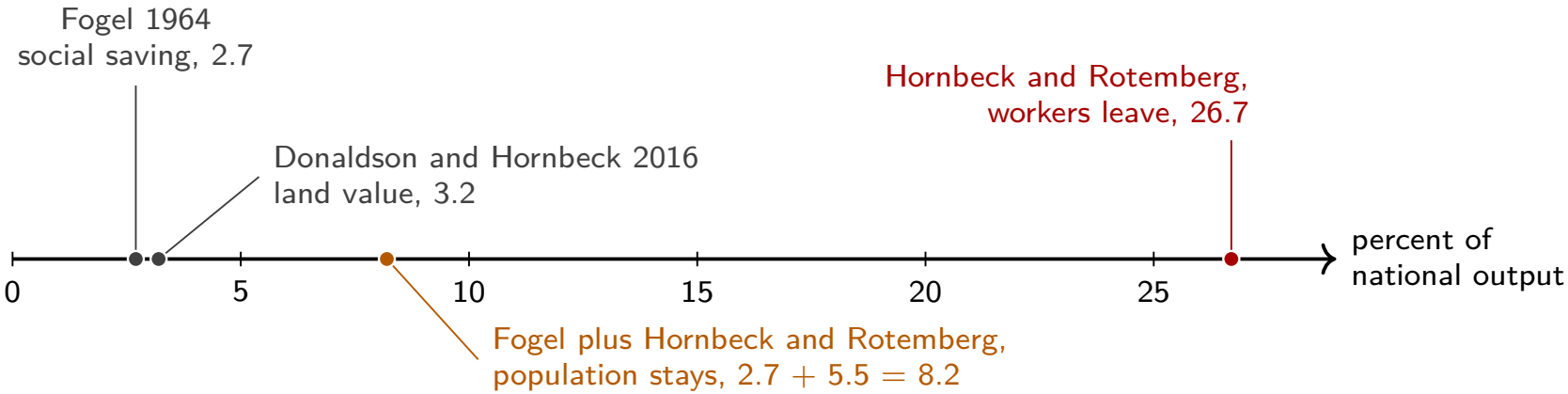
Where the AE came from	
Materials	0.107 (0.049)
Labor	0.066 (0.015)
Capital	-0.004 (0.009)

How the county grew	
Establishments	0.172 (0.034)
Revenue per establishment	0.021 (0.030)
Workers per establishment	0.024 (0.043)
Manufacturing share	0.0076 (0.0078)

The gaps, wedges, and cost shares did not change. Market access "did not make counties more efficient; rather, this encouraged the expansion of economic activity in otherwise distorted counties." More establishments of the same size opened and farming kept its share.

Hornbeck and Rotemberg (2024), Tables 3 and 5, pp. 3565 and 3568, and p. 3549. Manufacturing share is of county revenue.

Four railroad estimates measure different things



- Hornbeck and Rotemberg remove the 1890 railroads from a model of 2,722 counties. If workers leave until those who stay are as well off as before, the population falls 66 percent and productivity 26.7 percent. If the population stays, productivity falls 5.5 percent.
- Fogel counts transport costs and Donaldson and Hornbeck count land values. The two overlap since part of Fogel's figure is land that would have been abandoned.
- Hornbeck and Rotemberg's own model, wedges included, also gives a land-value loss of 3.5 percent of GNP. Their 5.5 and 26.7 are surplus that land prices do not capture. With no wedges they would be zero.

To the room. Which of the two would you report as the value of the railroads?

Fogel and Donaldson and Hornbeck report shares of GNP. Hornbeck and Rotemberg treat a percent of aggregate productivity as a percent of GDP. Hornbeck and Rotemberg (2024), pp. 3550 and 3583–3591, Table 9, and n. 1.

▸ Table 9

The social return to the railroads

- In the scenario where workers leave, a 26.7 percent loss in aggregate productivity is worth $3.2 billion a year in 1890 dollars. The whole network had cost about $8 billion.
- Adding the owners' earnings and the gain in land value gives a social return of 48 percent a year. The owners earned 3.5 percent, about 7 percent of the social return.
- Manufacturing productivity grew 2.2 percent a year from 1860 to 1900 so the loss equals about 12 years of that growth.
- Most of the gap between the two returns on the opening slide is the wedge. Investors were paid for the freight they carried and nothing for the inputs that expanded in every county they reached.

Hornbeck and Rotemberg (2024), pp. 3582–3585 and n. 17.

▸ The arithmetic

What is a public investment worth?

The question from the start was what an investment like Troesken's waterworks is worth to the economy as a whole.

	Investment	**Why the market price misses value**
Hornbeck and Keniston	a block of new buildings	each building raised its neighbors' rents and no owner was paid for it
Hornbeck and Rotemberg	a transport network	inputs already earned more than they cost so expanding them created surplus

When an investment expands activity that was too small, the price it fetches is a lower bound on its value. An externality and a wedge are two reasons activity is too small. Both papers end in a model and the model turns a local estimate into a total for a city or a nation.

Policy questions

- **Rebuilding.** If uncoordinated rebuilding cost a growing city about as much as its buildings were worth, what should a city do short of a fire? Land assembly, redevelopment districts, and eminent domain each coordinate owners and each has been abused.
- **Subsidy.** If the railroads returned 48 percent to society and 3.5 to their owners, what does that imply about the land grants and subsidies they received?
- **Today.** Where are wedges large now, and which infrastructure would raise output most where they are?

Next week

- **Week 10, 9 November. The Legacy of Slavery.**
- Wright (1999) on the Civil Rights revolution as economic history.
- Althoff and Reichardt (2024) on Jim Crow and Black economic progress after slavery.
- Abramitzky et al. (2021) on linking historical census records, with the record-linkage module.

Hornbeck and Rotemberg's Southern labor wedges jumped in 1870, the first census after emancipation.

Remove the railroads of 1890

The national number comes from a model of trade among 2,722 counties in which workers move freely, capital flows in, each county's land is fixed, and wedges act as taxes paid to local owners.

Change in productivity (%)	No rails	1860 rails	Canals, no rails	Rails cost 2×
Workers leave	−26.7	−15.6	−23.5	−8.7
Population stays	−5.5	−4.1	−4.4	−1.4
well-being (utility)	−32.7	−18.3	−29.0	−11.1

- Which row is right depends on where people go. In the first workers leave until those who stay are as well off as before, so the population is 66 percent lower. This is the scenario behind the 48 percent return on the opening slide. In the second everyone stays and is poorer.
- The paper adds the 5.5 to Fogel's 2.7, because its gains come on top of the transport savings, and calls the total "roughly triple" Fogel. Its Table 9 calls the 26.7 the baseline.
- Fogel's proposed canals would have recovered 12 percent of the loss. Railroads at twice the price would still have delivered two-thirds of the gain.

Hornbeck and Rotemberg (2024), Table 9, p. 3584, and pp. 3550 and 3585–3586. Estimated on 1890 data, the model reproduces the 1860–1880 county estimate without being fit to it, 0.197 against the regression's 0.204 (Table 8). Aggregation follows Hulten (1978) with the wedge term of Baqaee and Farhi (2020), equations (17)–(18).

Back

Productivity growth splits into technology and allocation

The paper measures a county's productivity by its surplus, revenue minus spending on inputs, $Pr = R - \sum_k E^k$, with $s^k = E^k/R$ input k's share of revenue. Write $d\ln X$ for the percent change in X. A small change in surplus is $dPr = dR - \sum_k dE^k$. Divide by Pr and write each change as a percent of its own base,

$$d\ln Pr = \frac{R}{Pr}\, d\ln R - \sum_k \frac{E^k}{Pr}\, d\ln E^k = \nu\Big[d\ln R - \textstyle\sum_k s^k\, d\ln E^k\Big], \qquad \nu = R/Pr,$$

since $E^k/Pr = s^k\nu$. Add and subtract $\nu \sum_k \alpha^k\, d\ln E^k$.

$$d\ln Pr \;=\; \underbrace{\nu\Big[d\ln R - \textstyle\sum_k \alpha^k\, d\ln E^k\Big]}_{\text{TFPR}} \;+\; \underbrace{\nu\Big[\textstyle\sum_k (\alpha^k - s^k)\, d\ln E^k\Big]}_{\text{allocative efficiency}}$$

- **TFPR.** Revenue grows by α^k percent for every 1 percent more of input k, so revenue growth beyond $\sum_k \alpha^k d\ln E^k$ is growth in revenue per unit of inputs, Week 5's TFPR. It is the whole story when every $\alpha^k = s^k$.
- **Allocative efficiency** (AE) is input growth weighted by the gaps. Each 1 percent more of input k raises revenue by α^k percent of R and costs s^k percent of R.
- $\nu \approx 5$ because surplus is about a fifth of revenue, so each percent of revenue gained with inputs fixed is 5 percent of surplus.

Hornbeck and Rotemberg (2024), equations (2)–(3), p. 3559. ν is 5.1 in the data (n. 6).

Back

How a 26.7 percent loss becomes a 48 percent return

The paper treats the loss in aggregate productivity as a share of 1890 GDP, about $12.0 billion, so 26.7 percent is $3.2 billion a year.

Annual return, millions of 1890 dollars	
Private return to the railroads	285
Annualized rise in farmland value	414
Gain in aggregate productivity, $0.267 \times 12{,}000$	3,204
Social return	3,903
Cost of the network, including equipment and land	8,121

- Social return $= 3{,}903/8{,}121 = 48$ percent a year.
- Private return $= 285/8{,}121 = 3.5$ percent a year.
- Investors' share $= 285/3{,}903 = 7$ percent of the social return.
- The $3,204 million comes from the scenario in which workers leave. With the population fixed the loss is 5.5 percent of GDP, $660 million, and the same sum gives $1{,}359/8{,}121 = 17$ percent, our calculation holding the other two terms at the paper's values.
- The returns are one year's flow and the cost is the 1890 stock, so 48 percent is a one-year return on what the network had cost by 1890.

The private return is net income 145, plus debt interest 217, plus net capital expenditure 5, less losses 30 and other income 52. The cost is 8,041 for track and equipment plus 80 for land. Hornbeck and Rotemberg (2024), p. 3585 and n. 17, from Adams (1895).

◂ Opening ◂ Social return